IFA BORU
SPECIAL BOARDING SECONDARY SCHOOL

Nuclear Physics
Worksheet
Prepared For Summer Program

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1. The half-life of radium-224 is 3.5 days. What fraction of the sample remains undecided after fourteen days? (if needed use: $\ln 2 = -\ln(1/2) = 0.693$)

   A) $\frac{1}{2}$ B) $\frac{1}{4}$ C) $\frac{1}{8}$ D) $\frac{1}{16}$

2. When the radioactive element $^{144}_{60}Nd$ decays to $^{140}_{58}Ce$, which one of the following particles is emitted?

   A) A beta particle C) a proton
   B) An alpha particle D) a gamma ray

3. The activity of a radioactive source falls to one sixteenth its original value in 32 minutes. What is the half-life, in minutes, of this decay process?

   A) 2 min B) 8 min C) 4 min D) 16min

4. Which statement is NOT correct about the emission from radioactive isotope?

   A) Alpha particles are negatively charged and more penetrating than beta and gamma.
   B) Beta particles are negatively charged and more penetrating than alpha particles.
   C) Alpha emissions are least influenced by magnetic field.
   D) The most penetrating emissions are gamma particles.

5. A common unit of energy in atomic physics is:

   A) Plank's constant C) de Broglie wavelength
   B) Electron volt D) uncertainty principle

6. The activity of radioactive substances reduces from 2000Bq to 1000Bq in 10^6 seconds. What is the decay constant?

   A) $6.93 \times 10^5 decay/s$ C) $2.00 \times 10^6 decay/s$
   B) $6.93 \times 10^{-7} decay/s$ D) $5.00 \times 10^{-7} decay/s$

7. Which of the following phenomena clearly demonstrates the particle nature of light?

   A) Refraction of light by prism C) Diffraction of light by slit
   B) The photoelectric effect D) Dispersion of white light

8. Which of the following has an electron volt unit?

   A) Charge C. Potential difference
   B) Energy D. Electromagnetic intensity

9. In Rutherford experiment, a beam of positively charged alpha particles was projected in to a thin piece of a gold foil. One of the observation of this experiment is that some charged particles deflected through a large angel (10^0 to 90^0) are those concentrated in one place. What conclusion can be drawn from this observation?

   A) Most of the mass and all positive charge is in tiny central nucleus
   B) Most of the atom is empty space
   C) All the atom is positive charge

D) All the atom is negative charge

10. Calculate the ratio of volume of uranium-238 isotope to the volume of its Nucleus.

A) 238 B) $\frac{1}{1770}$ C) 1770 D) $\frac{1}{238}$

11. Which, among the following are isotopes?

A) ${}^{a}_{b}X$; ${}^{b}_{c}Y$ B) ${}^{a}_{b}X$; ${}^{a}_{b+1}Y$ C) ${}^{a}_{b}X$; ${}^{a+1}_{b}Y$ D) ${}^{a}_{b}X^{a-1}_{b-1}Y$

12. Find the binding energy per nucleon in carbon-14 nucleus (${}^{14}_{6}C = 14u$, $m_p = 1.00728u$, $m_n = 1.00867u$).

A) 7.84MeV B) 7.52MeV C) 8.82MeV D) 7.60MeV

13. What the nuclear reaction, ${}^{a}_{b}X \rightarrow {}^{a}_{b-1}Y + {}^{0}_{+1}Z$, represents?

A) Electron emission C) Positron emission
B) Neutron emission D) proton emissio

14. Injected radioactive sources can be used as tracers to make soft tissues, such as blood vessels or the kidneys, show up through medical imaging processes. Which radiation type should the radioactive source used emit?

A) Alpha Particles B) Beta Particles C) Electrons D) Gamma Particles

15. Which of the following principle is not advised to minimize the radiation exposure?

A) Keeping distance from the radiation source
B) Bravely working with radiation without hesitation.
C) Not to stay in radiation area for long time
D) Using the radiation shielding

16. In the fusion reaction below determine the energy released. $(m({}^{1}_{1}H) = 1.00825u, m({}^{4}_{2}He) = 4.0026u)$

$$4{}^{1}_{1}H \rightarrow {}^{4}_{2}He + 2{}^{0}_{+1}\beta + Energy$$

A) 3726.82MeV B) 3755.10MeV C) 26.72MeV D) 7481.90MeV

17. In the radioactive decay of radon-222, both alpha and gamma radiation are emitted, with the latter having energy of 8.2×10^{14} J per nucleus decayed. How much energy does 2mol of Rn release in the form of gamma radiation?

$$^{222}_{86}Rn \rightarrow {}^{218}_{84}Po + {}^{4}_{2}\alpha + {}^{0}_{0}\gamma$$

A) $3.64 \times 101^{7}J$ B) $1.82 \times 10^{17}J$ C) $4.93 \times 10^{38}J$ D) $9.87 \times 10^{38}J$